Nursadul Mamun

7650, McCallum Blvd, Apt #1602
Dallas, TX-75252, USA
Mobile: (+1) 682-365-5363
Email: nursadul.mamun@utdallas.edu | nursad49@gmail.com

Authorized to work (F-1 Visa) in USA
Cochlear Implant Processing Laboratory,
Centre for Robust Speech Processing (CRSS)
ECSN 4.418
The University of Texas at Dallas,
800 West Campbell Road, Richardson, TX-USA
https://www.linkedin.com/in/nursadul-mamun-85980752/

OBJECTIVE: "Seeking an internship opportunity to contribute to a dynamic team in the field of signal processing, audio/speech/hearing, machine learning."

RESEARCH INTEREST:

- Speech Enhancement in Hearing Aid and Cochlear Implants
- Speech & Speaker Recognition/verification
- Voice Conversion
- Machine Learning (DNN, CNN, RNN)
- Speech Intelligibility and Quality Measure

WORK EXPERIENCE:

RESEARCH ASSISTANT
Cochlear Implant Processing Laboratory (Current, from May 2018)
University of Texas at Dallas
Research Assistant

Since 2018, I am extensively focusing on speech enhancement (SE) algorithm, capability of speaker identification, and intelligibility/quality metric for CI users. We propose a CNN-based speech enhancement algorithm for CI recipients performing in a cochlear filter-bank feature space. We also propose an algorithm to quantify CI users' ability for speaker identification. These works are accepted in InterSpeech and CIAP conference (2019) for publication. I am also working on voice conversion for the CI users to improve user's speech intelligibility.

Dept. of Electronics and Telecommunication Engineering, (April 2016 – May 2018)
Chittagong University of Engineering and Technology (CUET), Bangladesh
Assistant Professor

Auditory Neuroscience Laboratory
University of Malaya, Malaysia (Jan 2013 – Dec 2015)
Research Assistant

PROJECTS:

- Auditory features-based CNN Speech Enhancement algorithm for cochlear Implant recipients.
- Self-supervised CNN structure for speech de-noising under naturalistic environment.
- Objective speech Intelligibility metric for normal/cochlear-implant users.
- Voice conversion system for hearing benefits for hearing impaired subjects.
- Machine learning-based speaker identification using auditory features.

EDUCATION

Ph.D. in Electrical and Electronic Engineering
University of Texas at Dallas, Richardson, Texas, USA. May 2018 to Present (Pursuing)
Focus: *Speech Enhancement for Cochlear Implant users.*
Advisor: Professor John H. L. Hansen (email: john.hansen@utdallas.edu)

Master of Science in Biomedical Engineering Graduated: July 2015
University of Malaya, Kuala Lumpur, Malaysia. 'DISTINCTION' (Fully Research mode).
Focus: *Speech intelligibility prediction measure.*
Advisor: Dr. Md Shamsul Arefeen Zilany (email: msazilany@gmail.com)

Bachelor of Science in Electrical and Electronic Engineering Graduated: December 2012
Chittagong University of Engineering and Technology, Bangladesh. (3.89 with 'HONORS' out of 4; 90.76 %)

Completed/Ongoing Courses

- Digital Signal Processing
- Machine Learning
- Pattern Recognition and Machine Learning
- Speech and Speaker Identification
- Speech Perception
- Probability, Statistics, and Random variable

SKILLS AND ABILITIES

- **Programming:** MATLAB, Python, C, Simulink.
- **Tools:** Visual C/C++ Studio, SPROKITS, KALDI.
- **Environment:** Windows, Linux.

SELECTED PUBLICATIONS:

Journal Papers:

- **Nursadul Mamun**, W. A. Jassim, S. A. Zilany. "Prediction of Speech Intelligibility Using a Neurogram Orthogonal Polynomial Measure (NOPM)"). IEEE Transaction on Audio, Speech, and Language Processing, 2015 (ISI/SCOPUS *Cited Publication*), vol. 23(4), pp. 760-773.
- **Nursadul Mamun**, S. A. Zilany. "An intrusive method for estimating speech intelligibility from noisy and distorted signals". (Submitted)

Conference Papers:

- **Nursadul Mamun,** Soheil Khorram, John H.L. Hansen "Convolutional Neural Network-based Speech Enhancement for Cochlear Implant Recipients, INTERSPEECH 2019, 15-19 September 2019, Graz, Austria**.**
- **Nursadul Mamun,** Ria Ghosh, John H.L. Hansen "Quantifying Cochlear Implant Users' Ability for Speaker Identification using CI Auditory Stimuli, INTERSPEECH 2019, 15-19 September 2019, Graz, Austria**.**
- Khadija Akter**, Nursadul Mamun**, "Predicting Speech Intelligibility with the Regeneration of Envelope from TFS Cues for Hearing Impaired Listeners" EECE**,** 7-11 February 2019, Bangladesh.
- Md. Ibrahim Khalil, **Nursadul Mamun**, Khadija Akter "Robust Text Dependent Speaker Identification Using Neural Responses from the Model of the Auditory System" EECE**, 7-11** February 2019, Bangladesh.
- John H.L. Hansen, Hussnain Ali, Juliana N. Saba, Ram Charan M. C., **Nursadul Mamun**, Ria Ghosh, Avamarie Brueggeman "CCi-MOBILE: Design and Evaluation of a Cochlear Implant and Hearing Aid Research Platform for Speech Scientists and Engineers" IEEE-EMBS International Conference on Biomedical and Health Information, University of Illinois at Chicago, Chicago, IL,USA**.** (accepted)
- **Nursadul Mamun**, W. A. Jassim, S. A. Zilany. "Robust Gender Classification using Neural responses from the Model of the Auditory System" IFESS**,** 17-19 September 2014, Malaysia.
- **Nursadul Mamun**, W. A. Jassim, S. A. Zilany. "Speech-based Gender Classification Using Neurogram Orthogonal Polynomial measure**"** RCEE, 04-05 Mar 2014, Malaysia.
- Shoumya Chowdhury, **Nursadul Mamun**, A. A. Shahjamal Khan, Fahim Ahmed. "Text Dependent and Independent Speaker Recognition Using Neural Responses from the Model of the Auditory System" EECE-2016, 11-13 February 2017, Bangladesh.
- Khadija Akter**, Nursadul Mamun** "*Prediction of Speech Perception with Recovered Envelope Cues from TFS stimulus For Normal Hearing Listener*" ICEEICT 2018, 13-15 September 2018, MIST, Dhaka, Bangladesh**.**
- Taieba Athay, **Nursadul Mamun**, "Predicting Speech Intelligibility Based on Neural Cross-Correlation for Normal Hearing People" 10th International Conference on Electrical and Computer Engineering, 20-22 December 2018, Dhaka, Bangladesh**.**
- Md. Saiful Islam**, Nursadul Mamun** and Muhammad S. Ullah "Speech Based Deception Detection Using Bispectral Analysis", ICGSP, 2018, Australia**.**

Conference Abstracts or Presentations:

- **Nursadul Mamun**, Soheil Khorram, John H.L. Hansen " CCi-MOBILE: Environment-specific speech enhancement with cochlear implant listeners using convolutional neural network, 2019 Conference on Implantable Auditory Prostheses (CIAP), Lake Tahoe, California, USA**. (accepted)**
- Ria Ghosh, Juliana N. Saba**, Nursadul Mamun**, Hussnain Ali, John H.L. Hansen "CCi-Mobile: moving towards exploring advanced research paradigms for cochlear implant and hearing aid users, 2019 Conference on Implantable Auditory Prostheses (CIAP), Lake Tahoe, California, USA**. (accepted)**
- Evelyn E. Davies-Venn , **Nursadul Mamun** , Md Hossain , Timothy Kwan, Melanie Putman , MSA Zilany "Comparison of the predictive accuracy of different computational models of auditory perception" 177th Meeting Acoustical Society of America(ASA), 13-17 May 2019, Louisville, Kentucky**, USA.**
- **Nursadul Mamun**, Khadija Akter, Hussnain Ali, John H. L. Hansen "*Measuring Speech Perception with Recovered Envelope Cues using the Peripheral Auditory Model*" 176th Meeting Acoustical Society of America(ASA), 5-9 November 2018, Victoria, Canada**.**
- Hussnain Ali, **Nursadul Mamun**, John H. L. Hansen, "The CCi-MOBILE VOCODER" 176th ASA, 5-9 November 2018, Victoria, Canada**.**
- S. A. Zilany**, Nursadul Mamun**, W. A. Jassim. "Prediction of Behavioral Speech Intelligibility and Quality using a Computational Model of the Auditory System" 37th Mid Winter Meeting of the Association for Research in Otolaryngology (ARO), 21-25 Feb 2014,
- Evelyn Davies-Venn**, Nursadul Mamun**, S. A. Zilany. "Computational auditory model of intensity effects on amplified speech perception" 39th Mid-Winter Meeting of the Association for Research in Otolaryngology (ARO), 20-24 Feb 2016, San Diego, California, USA.

REFERENCES:

John H.L. Hansen, Ph. D.
Associate Dean for Research, Erik Jonsson School of Engineering and Computer Science
Professor, Dept. of Electrical Engineering, Erik Jonsson School
Center for Robust Speech Systems, Coordinator
The University of Texas at Dallas
email: John.Hansen@utdallas.edu; web: http://crss.utdallas.edu